

WipeDeploy

NIST SP 800-88 Sanitization & Windows Reinstall System

Electronics recycler edition – WiPE-based, USB-deployable, minimal technician interaction.

Standard: NIST SP 800-88 Rev.1 Media: HDD / SATA SSD / NVMe Levels: Clear & Purge

OS: Windows 10/11 Unattended

OVERVIEW

How the System Works

A technician inserts the deployment USB into the target machine and boots from it. WiPE loads automatically, finds the USB, and launches the deployment script. The technician answers two prompts – sanitization level and their ID – then walks away. The system detects the drive type, executes the correct NIST-compliant wipe, extracts the machine's OEM product key from UEFI firmware, partitions the disk, and launches an unattended Windows installation. A timestamped log is written to the USB at the end of every run.

01

Boot from USB into WinPE

STARTNET.CMD executes automatically. wpeinit loads drivers. Script USB is located by scanning drive letters.

02

TechNICIAN input

Only interaction

Select Clear or Purge. Enter techNICIAN ID. Script proceeds automatically from this point forward.

03

OEM Product Key Extraction

PowerShell reads the OA3 key from UEFI ACPI/MSDM table. Logged as UEFI_MSDM or NOT_IN_FIRMWARE. Windows Setup uses the key automatically if present.

04

Drive Type Detection

WMIC and PowerShell query disk 0. Model string and MediaType determine: NVMe / SATA SSD / HDD. Result logged.

05

NIST SP 800-88 Sanitization

Appropriate method executed based on drive type × sanitization level. Exit code checked. Non-zero exits halt the process — machine is not redeployed.

06

Partition & Unattended Install

GPT layout created (EFI 260 MB / MSR 16 MB / Primary NTFS). auto-unattended.xml generated with extracted key. setup.exe launched silently.

07

Sanitization Record Write

Log file saved to USB:\Logs\[SERIAL]_[TIMESTAMP].log. Contains machine info, drive details, method used, NIST level, exit codes, and final outcome.

Tools to Download

The following tools must be placed in the `Tools\` folder of the USB before the system is used. All are free and redistributable for commercial use.

`sdelete64.exe` – Microsoft Sysinternals

REQUIRED

Performs logical overwrite passes of HDD and SATA SSD volumes. Used for NIST Clear (1-pass) and NIST Purge (3-pass) of rotating media. Free, no license required for commercial use.

Download: <https://learn.microsoft.com/en-us/sysinternals/downloads/sdelete>

`nvme.exe` – `nvme-cli` Windows Port

REQUIRED

Sends NVMe format and sanitize commands directly to NVMe controllers. Required for `--ses=1` (User Data Erase / Clear) and `--ses=2` (Cryptographic Erase / Purge). The script attempts `ses=2` first and falls back to `ses=1` if the drive does not support it – the fallback is logged.

Download: <https://github.com/linux-nvme/nvme-cli> – use the pre-built Windows binary from the releases page.

Windows ADK + WinPE Add-on

REQUIRED

Needed to build the WinPE boot image and inject STARTNET.CMD. Install the ADK for your target Windows version. The WinPE add-on is a separate download of the same page.

Download: <https://learn.microsoft.com/en-us/windows-hardware/get-started/adk-install>

Windows 11 ISO

REQUIRED

Downloaded from Microsoft's Media Creation Tool or VLSC. Mount the ISO and point

`BUILD_USB.bat` at the mount point. The Windows source files (~4–5 GB) are copied to the

USB.

Download: <https://www.microsoft.com/en-us/software-download/windows11>

USB SIZE REQUIREMENT

Windows source files are approximately 4–5 GB. Combined with WinPE and tools, a 16 GB USB is the minimum. 32 GB is recommended to allow log accumulation over time without clearing the drive.

FILE LAYOUT

USB Drive Structure

The USB must be formatted NTFS (not FAT32 — Windows source files exceed the FAT32 4 GB file size limit). Run `BUILD_USB.bat` to populate the structure automatically.

```
USB:\ |— WIPE_AND_DEPLOY.bat ← Main script. Do not rename. |—
BUILD_USB.bat ← Run once on workstation to populate USB |— VIEW_LOGS.bat ←
Log viewer / report summary |— STARTNET.CMD ← Injected into WinPE image
(not on USB root) |— Tools\ |— sdelete64.exe ← Sysinternals |—
nvme.exe ← nvme-cli Windows port |— Windows\ ← Copied from Windows ISO |
|— setup.exe | |— sources\ | |— boot\ | |— ... |— Logs\ ← Created
automatically. One .log per machine. | |— [SERIAL]_[TIMESTAMP].log |—
Work\ ← Runtime temp. autounattend.xml written here.
```

COMPLIANCE REFERENCE

NIST SP 800-88 Rev.1 — Method Mapping

The table below documents exactly which tool and command the script executes for each combination of drive type and sanitization level, and the corresponding NIST SP 800-88 Rev.1 section.

DRIVE				NIST
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TYPE	LEVEL	TOOL	COMMAND / METHOD	REF.
HDD	Clear	sdelete64	1-pass overwrite with zeros (<code>-p 1 -z</code>)	Sec 2.3 / Table A-1
HDD	Purge	sdelete64	3-pass overwrite (<code>-p 3 -z</code>). See Note ①	Sec 2.4 / Table A-1
SATA SSD	Clear	diskpart	Full-disk zero write (<code>clean all</code>)	Sec 2.3 / Table A-1
SATA SSD	Purge	diskpart	<code>clean all</code> + Purge-attempted notation. See Note ②	Sec 2.4 / Table A-1
NVMe	Clear	nvme.exe	User Data Erase (<code>format --ses=1</code>)	Sec 2.3 / Table A-1
NVMe	Purge	nvme.exe	Cryptographic Erase (<code>format --ses=2</code>), fallback to ses=1 if unsupported. See Note ③	Sec 2.4 / Table A-1

NOTES

① HDD Purge: NIST 800-88 Rev.1 Sec 2.4 states that overwrite alone may not constitute Purge for all HDD types — degaussing or physical destruction provides higher assurance. The script performs 3-pass overwrite and logs it as Purge-attempted. If your clients require certified Purge for HDDs, degauss or shred the drive and document it separately.

② SATA SSD Purge: True NIST Purge for SATA SSDs requires ATA Secure Erase Enabled via a tool capable of sending ATA security commands (hdparm on Linux, or a manufacturer utility). `diskpart clean all` is a full zero-write and satisfies Clear. The script logs Purge-attempted to flag this gap for your records.

③ NVMe ses=2 Fallback: Not all NVMe drives support Cryptographic Erase. When ses=2 fails, the script automatically retries ses=1 and records the fallback in the log. Review fallback entries before releasing machines processed for HIPAA or DoD clients.

ONE-TIME SETUP

Building the WinPE Boot Image

This is performed once. After the WinPE image is built and written to the USB, it does not need to be rebuilt unless you update STARTNET.CMD or add WinPE packages.

Step 1 — Create the WinPE working environment

```
:: Run from the Windows ADK deployment tools command prompt (as Administrator)
cotype amd64 C:\WinPE_amd64
```

Step 2 — Mount the WinPE WIM

```
dism /mount-wim /wimfile:C:\WinPE_amd64\media\sources\boot.wim ^
    /index:1 /mountdir:C:\WinPE_amd64\mount
```

Step 3 — Inject STARTNET.CMD

```
copy /y STARTNET.CMD C:\WinPE_amd64\mount\Windows\System32\startnet.cmd
```

Step 4 — Add WMI support (required for drive detection)

```
dism /image:C:\WinPE_amd64\mount ^
    /add-package /packagepath:"C:\Program Files (x86)\Windows Kits\10\Assessment

dism /image:C:\WinPE_amd64\mount ^
    /add-package /packagepath:"C:\Program Files (x86)\Windows Kits\10\Assessment
```

Step 5 — Unmount and commit

```
dism /unmount-wim /mountdir:C:\WinPE_amd64\mount /commit
```

Step 6 — Write WinPE to USB

```
:: Replace E: with your USB drive letter
MakeWinPEMedia /UFI C:\WinPE_amd64 E:
```

AFTER THIS STEP

Run `BUILD_USB.bat` to copy the scripts, tools, and Windows source to the USB. MakeWinPEMedia formats the USB and writes the boot environment — BUILD_USB.bat then populates the data partition.

RECORD KEEPING

Sanitization Log Format

Every machine produces a single log file at `USB:\Logs\[BIOS_SERIAL]_[TIMESTAMP].log`. The file is plain text and contains a complete sanitization record suitable for audit purposes.

Sample Log — Pass

AUTO-GENERATED

```
=====
NIST SP 800-88 Rev.1 SANITIZATION RECORD
=====

Script Version : 1.0
Standard       : NIST SP 800-88 Rev.1
Date/Time      : 2025-06-04_143022
Manufacturer   : Dell Inc.
Model          : OptiPlex 7090
BIOS Serial    : 8XKPT23
-----

Sanitization Level : PURGE (operator selected)
Technician ID      : JM
-----

[STEP 1] Product key extraction
Key source : UEFI_MSDM
```

```
Key          : XXXXX-XXXXX-XXXXX-XXXXX-XXXXX
[STEP 2] Drive detection
Drive type   : NVMe
Drive ID     : 0
Drive size   : 512 GB
Model        : Samsung MZVLB512HAJQ-000H1
[STEP 3] Sanitization begin
Method       : NVMe-PURGE: nvme format --ses=2 Cryptographic Erase
Sanitization exit code : 0
[STEP 3] Sanitization PASSED (exit code 0)
[STEP 4] Partitioning complete
[STEP 5] autounattend.xml written
[STEP 6] setup.exe launched successfully (exit 0)

-----

FINAL OUTCOME
Sanitization : PASS
OS Install   : LAUNCHED (exit 0)
Record saved : E:\Logs\8XKPT23_2025-06-04_143022.log
=====
```

OPERATIONAL NOTES

Known Limitations

SATA SSD True Purge

LIMITATION

The script cannot issue ATA Secure Erase executed natively from WinPE without an additional tool such as a manufacturer's utility (e.g., Samsung Magician CLI, Crucial Storage Executive CLI). If true NIST Purge of SATA SSDs is required for your client contracts, the recommended path is to add a Linux-based WinPE alternative (such as a Parted Magic boot USB run before this one) or source and add a manufacturer CLI to the Tools folder and extend the SSD Purge branch of the script accordingly.

Retail / Volume License Keys

LIMITATION

The key extraction reads only from the UEFI ACPI/MSDM table. OEM machines (Dell, HP, Lenovo factory installs) almost universally have the key here. Machines with retail box licenses or volume licenses stored in software will return NOT_IN_FIRMWARE. Windows Setup will still proceed — it will simply require activation after install. These machines are flagged clearly in the log.

Multi-Disk Systems

LIMITATION

The script targets disk 0 only. Machines with secondary drives (a second HDD in a workstation, for example) will have those drives untouched. If your intake includes multi-drive systems, the sanitization loop should be extended. This is a straightforward script modification — the subroutine structure is in place to support it.